

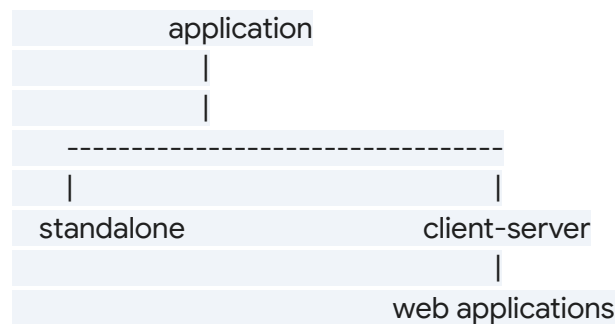
APPLICATION FUNDAMENTALS

Date of Lecture: March 9th

1. Application

- An Application can be considered as a set of codes and instructions which perform some specific tasks.
- An application works by taking an Input then operates on data to return an output.
- **Examples:** calculator, camera, Youtube, WhatsApp.

Application Hierarchy:



2. Standalone

- It is considered as an independent application which does not rely on external resources.
- It is also known as a "self functioning application."
- It is also known as a "single user application."
- **External resources include:** Internet, server, and database.
- **Examples:** calculator, camera, snake game.
- **Types based on Installation:** System application and User application.

3. Client Server Application

- **Client:** It is considered a "device" which is capable of having internet connections and installing an application.
- **Server:** A server is a big or powerful computer which is specifically designed in order to handle multiple users at the same time without data loss or data lag.
- The server contains the source code which processes the input and produces the result.

4. Web Applications

- It is a type of application which can only be accessed on a browser.
 - It is also considered as a collection of webpages.
 - Web applications work on the same principle of a client-server application.
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5. Browser

- It is considered as both a standalone and a client-server application.
 - **Examples:** chrome, edge, safari, brave, firefox, opera, onion.
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6. Key Differences: Traditional Client-Server vs. Web Applications

Feature	Traditional Client-Server Application	Web Application
Response Time	Faster: The user interface is installed locally, so it only needs to fetch data from the server, making interactions feel instantaneous.	Slower: Every action often requires loading both the user interface elements and the data over the internet.
Storage Perspective	Heavy Local Storage: Requires downloading and installing the full application on the user's device, consuming local hard drive space.	Best for Storage: Requires zero local installation space. The entire application is hosted remotely.
Platform Dependency	Platform Dependent: Must be specifically developed and	Platform Independent: Can run on any device as long as it

	installed for a particular Operating System.	has a compatible web browser.
Updates & Maintenance	Manual Updates: Every individual user must download and install updates on their specific device to get the latest version.	Centralized Updates: Updates are made once on the server. All users instantly see the updated version the next time they refresh.
Device Accessibility	Device Specific: You can only use the application on the exact device where you installed it.	Highly Accessible: You can access your account and use the application from anywhere, on any device, by logging in.
Resource Utilization	High Local Resource Use: Relies heavily on the client device's CPU and RAM to render graphics and process local logic.	Low Local Resource Use: The server handles the heavy processing; the client device only needs enough power to run a browser.
Security	Controlled Environment: Generally considered more secure as it operates in a closed environment with strict, custom local protocols.	Public Exposure: More vulnerable to common internet-based attacks since it is exposed over standard web protocols.
Internet Dependency	Partial: Sometimes allows limited offline use or local processing until a connection to the server is restored.	Complete: Absolutely requires an active internet connection to function.